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Rapid Magnitude Estimation for Taiwan Earthquake Early Warning via Frequency-Domain Multi-Station Deep Learning

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Keywords:

Earthquake Early Warning System (EEWS); Earthquake magnitude estimation; Machine learning

Abstract

Accurate and rapid earthquake magnitude estimation is essential for effective Earthquake Early Warning Systems (EEWS), particularly in seismically active regions such as Taiwan. In this study, we propose a frequency-domain multi-station deep learning approach that integrates a Fourier Neural Operator (FNO) and a Long Short-Term Memory (LSTM) network to estimate earthquake magnitudes using near real-time waveform data. The model is trained on 7,025 events recorded by Taiwan's networks between 2012 and 2024. By jointly analyzing three-component waveforms from the ten nearest stations and incorporating hypocentral distance corrections, the proposed method achieves high accuracy within seconds after the fourth triggered P-wave arrival. Evaluation on a held-out test set demonstrates significant improvements over the current Pd-based single-station method currently used by the Central Weather Administration (CWA), with lower mean absolute error, root mean squared error, and prediction variance. In real EEWS scenarios from January to June 2025, the model consistently outperformed the operational method, even when using early-stage location estimates, and showed further gains when provided with precise hypocenters from the catalog. The proposed approach combines robustness, and computational efficiency, enabling practical deployment in EEWS of Taiwan.

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1.Introduction

Since reliable earthquake prediction remains elusive, Earthquake Early Warning Systems (EEWSs) have emerged as a practical means of mitigating earthquake hazards by issuing alerts before the arrival of strong ground shaking at a given location (e.g., Gasparini et al. 2007;

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Allen and Melgar 2019). The fundamental principle of EEWSs is that electromagnetic communication signals propagate much faster than seismic waves, and that among seismic waves, P-waves travel faster than the more damaging S-waves. When a significant earthquake occurs, EEWSs are designed to rapidly detect the event and use the initial P-wave signals to estimate the hypocenter location and earthquake magnitude. If the predicted ground motion at a target site, based on these estimated source parameters, exceeds a predefined threshold, the system issues a warning to the public (e.g., Gasparini et al. 2007; Chen et al. 2015, 2019; Allen and Melgar 2019). The time interval between the issuance of the warning and the arrival of strong shaking, known as the lead time, provides critical seconds for individuals to take protective actions and for facilities to initiate automated safety measures.

A critical component of EEWS is the near real-time estimation of source parameters, among which earthquake magnitude is a key factor for timely decision-making. In conventional EEWS, rapid magnitude estimation typically relies on characteristics of initial P-wave recordings from single stations, such as maximum amplitude or characteristic frequency, which are empirically regressed to earthquake magnitude (e.g., Allen and Kanamori 2003; Wu and Zhao 2006; Gasparini et al. 2007; Wu et al. 2007; Wu and Kanamori 2008; Chen et al. 2015, 2017; Allen and Melgar 2019). These methods are computationally efficient and can provide timely magnitude estimates for ground-shaking assessment within EEWS. However, because the algorithms utilize only short time windows from individual stations, the resulting estimates are susceptible to biases arising from factors such as source mechanism variations or site effects. One of the main challenges in operational EEWS is the station-to-station variability in magnitude estimates, which can lead to certain stations producing values that deviate significantly from the network average. To enhance the robustness of the final magnitude estimate, these outliers are often identified and excluded, and the remaining values are averaged to obtain a more reliable network estimate (e.g., Chen et al. 2015, 2019). While this procedure can improve overall accuracy, it introduces a notable trade-off, as the exclusion of certain station data, although intended to reduce error, may inadvertently discard seismologically meaningful information.

In recent years, machine learning (ML) has made remarkable progress in earthquake magnitude estimation (e.g., Navarro-Rodríguez et al. 2025). Traditional magnitude estimation approaches typically rely on source parameters and empirical scaling relationships, whereas emerging machine-learning and deep-learning techniques enable the direct extraction of informative features from seismic waveforms, allowing for more efficient and potentially more accurate magnitude estimation. For example, multiple features extracted from the early-arriving portions of seismic waveforms have been used as inputs to traditional machine-learning–based methods, including Support Vector Machines (SVM; Zhu et al. 2021) and Extreme Gradient Boosting (XGBoost; Lara et al. 2023;

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Joshi et al. 2023), for rapid earthquake magnitude estimation. These approaches rely on predefined waveform features as inputs to machine-learning models, which may limit their ability to fully capture the complexity of raw seismic signals. To overcome this limitation, neural network–based approaches, such as convolutional neural networks (CNNs), have been developed to automatically learn representative features directly from seismic waveforms for magnitude estimation (e.g., Mousavi and Beroza 2020). In addition, incorporating spatial information, multi-station observations, and multiple feature representations has been shown to further improve the performance of earthquake magnitude estimation (e.g., van den Ende and Ampuero 2020; Zhang et al. 2021; Münchmeyer et al. 2021; Saad et al. 2022; Song et al. 2022; Hou et al. 2024). Collectively, these studies demonstrate that machine learning methods can reliably extract informative features from seismic waveforms for magnitude estimation, and that incorporating multi-station recordings can further enhance estimation accuracy and robustness. Such rapid and reliable magnitude assessments are essential components supporting effective EEWS.

Taiwan is situated at a convergent plate boundary where the Philippine Sea Plate subducts beneath and collides with the Eurasian Plate, resulting in an active arc–continent collision zone (e.g., Ho 1986; Teng 1990; Huang et al. 2000; Byrne et al. 2011). This complex tectonic environment has produced numerous active fault systems and high seismicity. Over the past decades, Taiwan has experienced multiple destructive earthquakes, and when combined with its high population density, the seismic hazard poses a significant threat to public safety. Consequently, there is a need for a fast and reliable EEWS to mitigate the potential for substantial property damage and loss of life. To address this need, Taiwan has developed and implemented an EEWS operated primarily by the Central Weather Administration (CWA) (e.g., Wu et al. 2007a; Chen et al. 2015, 2019). This regional warning system leverages a dense seismic network to detect the first-arriving P-waves, from which it rapidly estimates the earthquake's hypocenter and magnitude. Based on these source parameters, the system predicts ground shaking intensity. If the estimated magnitude and predicted shaking exceed predefined thresholds in a given county, a warning is issued to the public in that area (Chen et al. 2015).

In the current CWA-operated system, magnitude is estimated from the vertical component of peak ground displacement (Pd) measured within the first three seconds after P-wave arrival (e.g., Chen et al. 2015, 2019). The resulting magnitude, denoted M_{Pd}, is calculated using empirical relationships that vary by seismic sensor type and account for hypocentral distance. Magnitude estimation is performed independently at each station, and to improve robustness, only M_{Pd} values within one standard deviation of the dataset are retained before averaging across stations (Chen et al. 2015). However, in real-time operations, removing outliers may also eliminate valuable seismic information. A multi-station

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joint estimation approach that directly integrates waveform data across stations has the potential to improve the completeness, accuracy, and reliability of magnitude estimation. In this study, we propose a machine learning–based framework that leverages three-component waveform data from multiple seismic stations to jointly estimate earthquake magnitude in near real time. Our approach aims to produce more accurate and robust magnitude estimates, thereby enhancing decision-making efficiency and the overall effectiveness of Taiwan's EEWS.

2. Data Preparation and Dataset Partitioning

The effectiveness of an Earthquake Early Warning System (EEWS), particularly in providing sufficient lead time, is strongly influenced by the density of the seismic network. Taiwan's original EEWS utilized data from both the Central Weather Administration Seismic Network (CWASN) and the Broadband Array in Taiwan for Seismology (BATS), enabling near real-time warnings (Chen et al. 2015). To further enhance the system's efficiency and responsiveness, starting in 2018, stations from the Taiwan Strong Motion Instrumentation Program (TSMIP) (Liu et al. 1999) were progressively upgraded to support real-time data transmission (Figure 1a). Owing to the high station density of TSMIP, this integration has substantially improved the EEWS in terms of operational speed, efficiency, and overall reliability.

To develop a machine learning model capable of rapidly estimating the magnitude of moderate-to-large earthquakes in Taiwan, we utilized strong-motion seismic waveform data collected from several Taiwan seismic networks. These include the CWASN, TSMIP, and the BATS. Among these, strong-motion sensors constitute the most numerous and spatially comprehensive instrumentation, offering high-quality ground motion recordings. Given that our model is designed for magnitude estimation of moderate-to-large earthquakes, strong-motion data are particularly suitable due to their ability to capture higher-amplitude and spatially distributed recordings compared to broadband or short-period sensors.

The earthquake events used in this study were selected from the CWA's official earthquake catalog, covering the period from January 2012 to December 2024. We included events with local magnitudes (M_L) greater than 3.5, and located within the geographical bounds of 119.0°E to 123.0°E longitude and 20.0°N to 26.0°N latitude. The maximum focal depth of selected events reaches 312 km. To ensure sufficient waveform coverage for training, only events recorded by at least 10 strong-motion stations were retained. In total, 7,025 earthquake events met these criteria and were incorporated into the dataset (Figure 1b-d).

For efficient and near-real-time identification of P-wave arrivals, we adopted the RED-PAN (Real-Time Earthquake Detection and Phase-Picking with Multitask Attention Network) deep learning model (Liao et

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al. 2022) for seismic phase picking. RED-PAN enhances the capability of real-time earthquake phase picking and has been trained and validated using both the Stanford Earthquake Dataset (STEAD) (Mousavi et al. 2019) and seismic recordings from Taiwan(Liao et al. 2022). The model integrates goal-oriented data augmentation strategies, such as random temporal shifting of waveforms, to improve the robustness of phase detection in continuous seismic data. Notably, it can identify P-wave arrivals using only approximately the first 0.1 seconds following the P-wave onset, making it highly suitable for earthquake early warning (EEW) applications (Liao et al., 2022). In practical applications, RED-PAN typically achieves P-wave detection within 0.3 seconds after receiving the waveform input. Accordingly, all P-wave arrival times used in this study are derived from RED-PAN predictions, ensuring consistency and low-latency performance for real-time implementation.

The seismic waveform data used in this study were sampled at 100 Hz, consistent with the current configuration of seismic networks in Taiwan. Raw recordings, originally stored in instrument counts, were converted to physical ground acceleration units (m/s^2) using calibration factors provided by the CWA. This conversion ensures that the machine learning model can capture the physical relationships between waveform amplitude, waveform characteristics, and earthquake magnitude. To preserve the absolute amplitudes essential for magnitude estimation, no normalization was applied to the waveforms (e.g., Mousavi and Beroza 2020; Zhang et al. 2021; Hou et al. 2024). A detrending process was performed to remove baseline drift and minimize the impact of initial offsets on amplitude estimation. Additionally, a one-sided Hanning taper was applied to the 0.5 s interval preceding the P-wave pick for each waveform. This tapering ensures that the complete P-wave onset is retained within the analysis window while suppressing background noise prior to the arrival, thereby allowing the extracted features to predominantly represent seismic signals relevant for magnitude estimation. For each earthquake event, recordings from the 10 nearest strong-motion stations were selected. Each record was 20 seconds in length, yielding an input matrix consisting of three components (E–W, N–S, and vertical) for each of the ten stations.

The current EEWS operated by the CWA is a regional system that typically requires at least 4 initial P-wave arrivals to estimate the hypocenter of earthquake. For each station, the CWA EEWS verifies P-wave picks using a 1.0-second time window after the pick, applying amplitude and signal-to-noise ratio checks to ensure phase reliability (Chen et al. 2015). Waveform segments for this study were generated following a stepwise procedure. The end time of the first segment was set to 0.5 second after the P-wave arrival at the fourth triggered station, the earliest time at which the CWA EEWS can provide a hypocenter estimate. Subsequent segments were extended in 0.5-second increments, progressively incorporating additional waveform information from continuous recordings (Figure 2). To simulate the variability inherent in

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real-time data transmission, the end time of each waveform segment was further extended by a random offset between 0.0 and 0.5 seconds. This approach introduces temporal uncertainty, improving the model's robustness and generalization capability. The end time of the *n*-th segment can be expressed as:

$$\text{End time} = t_{P4} + n \times 0.5\text{s} + \text{random}(0.0, 0.5\ \text{s}),$$

where t_{P4} is the P-wave arrival time at the fourth triggered station. For stations that had not yet been triggered (i.e., P-wave not yet arrived), the waveform inputs were set to zero to prevent unrelated noise from influencing magnitude estimates. The windowing process was terminated if the P-wave arrival at the first triggered station occurred less than 0.5 seconds from the start of the waveform window (Figure 2). In such cases, only the segments generated prior to this point were retained.

To enhance the model's generalization capability, we employed a data augmentation strategy in which the order of stations in the input array was randomly permuted. This design simulates realistic operational conditions, where the station that first records the P-wave arrival is not necessarily the first to transmit its waveform data to CWA. By introducing this variability, the augmented dataset better reflects potential transmission delays and network asynchrony, thereby improving the model's robustness in real-time applications.

After constructing the dataset, earthquake events were divided into training, validation, and test sets in an 8:1:1 ratio (e.g., Bishop 2006; Goodfellow et al. 2016). Partitioning was performed at the event level to prevent waveforms from the same earthquake from appearing in multiple subsets, thereby avoiding data leakage. Because large earthquakes occur much less frequently than small ones, the dataset exhibits an inherently imbalanced magnitude distribution. To address this, earthquake events were categorized into four magnitude intervals: $3.5 \le M < 4.0$, $4.0 \le M < 5.0$, $5.0 \le M < 6.0$, and $M \ge 6.0$. Within each interval, events were randomly assigned to the training, validation, and test datasets according to the 8:1:1 ratio. This stratified sampling ensured a balanced magnitude distribution across all subsets.

3. Methodology

In this study, we aim to leverage near real-time waveform recordings from multiple seismic stations to enable rapid and accurate earthquake magnitude estimation following event confirmation within the framework of EEWS. Recent advances in machine learning have achieved notable success in a variety of seismological applications, including seismic phase picking (e.g., Zhu and Beroza 2018; Mousavi et al. 2020; Liao et al. 2021, 2022), earthquake magnitude estimation (e.g., Mousavi and Beroza 2020; Zhang et al. 2021; Saad et al. 2022; Hou et al. 2024), and seismic source-type classification (e.g., Seydoux et al. 2020; Jenkins et al. 2021;

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Wang et al. 2023). These techniques can automatically extract relevant features from seismic recordings and utilize them for predictive modeling. Building upon these advancements, we employ machine learning models to extract features from near real-time seismic waveforms, with the objective of generating more accurate and reliable magnitude estimates. Improved magnitude estimation accuracy is anticipated to enhance decision-making processes in subsequent EEWS stages, thereby increasing both the timeliness and reliability of earthquake warnings.

3.1 Overall Architecture

The machine learning architecture developed in this study is illustrated in Figure 3 and comprises three primary components: a Fourier Neural Operator (FNO) module (e.g., Li et al. 2021; Sun et al. 2023), a Long Short-Term Memory (LSTM) network (e.g., Hochreiter and Schmidhuber 1997; Gers et al. 2000), and fully connected (FC) layers. The model takes as input both multi-station seismic waveforms and their corresponding hypocentral distances. The multi-station waveform input is represented as a tensor with dimensions $10 \times 3 \times 2000$, corresponding to 10 seismic stations, three components per station, and 20-second waveform segments sampled at 100 Hz. The waveform data from multiple stations are first processed by the FNO layers, which extracts both spatial and frequency-domain features. In parallel, the hypocentral distances are processed by an FC layer that applies a linear transformation to generate station-specific correction factors. These correction factors are then multiplied element-wise with the corresponding waveform features. This operation emulates conventional magnitude estimation methods, in which amplitudes are corrected by a distance-dependent term to account for seismic attenuation effects. The resulting combined representation is then passed to the LSTM network, which models the temporal dependencies in the waveform data and learns features relevant to magnitude estimation. Finally, the output from the LSTM is fed into a fully connected layer that produces a single scalar value representing the estimated earthquake magnitude.

3.2Fourier Neural Operator

The Fourier Neural Operator (FNO) is employed in this study to extract seismic waveform features for earthquake magnitude estimation. Originally proposed by Li et al. (2021), the FNO is a machine learning framework designed to learn resolution-invariant mappings between infinite-dimensional function spaces. Initially developed for solving partial differential equations (PDEs), FNO has demonstrated strong performance across various physical science applications.

Given that the frequency content of seismic waveforms carries essential information for magnitude estimation, we adopt the FNO as the core architecture for feature extraction. Compared to conventional

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convolutional neural networks (CNNs), the FNO leverages spectral convolution layers that operate in the frequency domain, allowing global interactions across the entire input domain. This design effectively provides an unbounded receptive field, enabling the model to capture long-range dependencies and global patterns in the data more efficiently, which is particularly advantageous for modeling the complex and distributed features of seismic waveforms. The relationship between consecutive layers in the FNO can be expressed as:

$$v_{t+1}(x) = \sigma(W v_t(x) + F^{-1}[R \bullet F(v_t)](x))$$

Here, v_t denotes the input tensor to the layer; F and F^{-1} represent the Fourier transform and its inverse, respectively; R is a periodic function applied in the frequency domain; W is a weight matrix corresponding to a linear transformation; and σ is a non-linear activation function (Figure 3).

Each FNO layer processes data through two parallel paths. In the spectral path, the input is first transformed into the frequency domain via a Fast Fourier Transform (FFT). Within this domain, element-wise multiplication with R functions as a spectral filtering operation, attenuating high-frequency noise while preserving low-frequency components that are relevant to magnitude estimation. The filtered representation is then transformed back to the spatial (or temporal) domain using an inverse FFT. In the spatial path, a standard linear transformation W is applied directly to the input in the original domain. The outputs from these two paths are subsequently summed and passed through a nonlinear activation function, which in this study is the Gaussian Error Linear Unit (GELU). The FNO implementation used in this study is adapted from the open-source PhaseNO framework (Sun et al. 2023), with modifications to accommodate the characteristics of seismic waveform data and the requirements of magnitude estimation.

3.3Long Short-Term Memory (LSTM)

Previous studies on magnitude estimation using machine learning have demonstrated that preserving and leveraging the temporal dependencies embedded in seismic waveform features can substantially improve estimation accuracy (e.g., Mousavi and Beroza 2020; Hou et al. 2024). In particular, it has been shown that waveform data, when used without normalization preprocessing, can still be effectively processed by the Long Short-Term Memory (LSTM) architecture (e.g., Hochreiter and Schmidhuber 1997; Gers et al. 2000). The inherent design of LSTM networks enables them to robustly handle variations in amplitude relationships, thereby reducing the risk of any single feature disproportionately influencing the estimation. For these reasons, the LSTM is incorporated into our model architecture to enhance both the accuracy and the reliability of magnitude estimation.

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The LSTM is a specialized type of recurrent neural network (RNN) designed to overcome the vanishing and exploding gradient problems that frequently arise when modeling long sequences (e.g., Hochreiter and Schmidhuber 1997; Gers et al. 2000). LSTMs achieve this through the use of memory cells and gating mechanisms, which allow the model to capture long-term dependencies in sequential data while selectively filtering out irrelevant short-term fluctuations. An LSTM unit comprises three primary gates: the forget gate, input gate, and output gate. The sigmoid activation function, producing values between 0 and 1, regulates the flow of information through these gates. The forget gate determines the extent to which information from the previous cell state (long-term memory) is retained or discarded, scaling the memory content according to the sigmoid output. The input gate controls the degree to which new information from the current input (short-term memory) is incorporated into the cell state. The output gate regulates how much of the updated cell state is revealed as output at each time step, based on both the current input and the internal memory. Empirical evaluations have identified the forget gate as the most critical component of the LSTM architecture, followed by the input and output gates(Jozefowicz et al. 2015). This finding emphasizes the importance of selectively retaining relevant historical information for effective sequence modeling. This capability is particularly valuable for seismic waveform analysis, where both immediate and long-term temporal dependencies play a critical role in achieving accurate magnitude estimation.

3.4 Loss Function

The loss function is a fundamental component in training machine learning models, as it quantifies the discrepancy between model predictions and the ground truth (e.g., Bishop 2006; Goodfellow et al. 2016). The choice of loss function plays a decisive role in shaping the model's learning behavior and optimization trajectory. In this study, we adopt the Mean Squared Error (MSE) loss as the baseline, given its suitability for regression tasks. This choice aligns with conventional earthquake magnitude estimation, which is inherently a regression problem. The MSE is computationally efficient, straightforward to implement, and easily interpretable. Moreover, its sensitivity to larger errors naturally encourages the model to focus on improving predictions where discrepancies are greatest which is an important consideration in magnitude estimation. The MSE is defined as:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - y_i)^2$$

where n is the number of samples, y_i is the predicted magnitude of the i-th earthquake, and y_i is the corresponding ground truth.

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Given that large-magnitude earthquakes are disproportionately associated with severe ground shaking, infrastructural damage, and societal losses, their accurate estimation is of particular importance for earthquake early warning applications. However, such events occur far less frequently than small-to-moderate earthquakes, resulting in an inherently imbalanced magnitude distribution in the training dataset. To address this imbalance and to ensure that the model adequately learns from the limited number of high-magnitude events, we designed a magnitude-weighted loss function that adjusts each earthquake's contribution to the total loss according to its true magnitude:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} e^{0.5y_i-3} \bullet (y_i - y_i)^2$$

The weighting factor $e^{0.5y_i-3}$ is introduced to enhance the contribution of large-magnitude earthquakes to the overall loss during model training. Its exponential form enables a smooth and continuous scaling effect, where higher-magnitude events receive increasingly larger weights. This design reflects the real-world importance of large earthquakes, which typically cause more intense ground shaking and greater societal impact. Moreover, the weighting factor is intended to address a common challenge in seismic datasets, the imbalance in magnitude distribution. Without compensation, this imbalance can bias the learning process, leading the model to prioritize performance on low-magnitude cases at the expense of higher-magnitude predictions. Re-weighting the loss to compensate for imbalanced data is a common strategy in machine learning (e.g., Elkan 2001; He and Garcia 2009; Krawczyk 2016). By increasing the loss contribution from underrepresented large-magnitude events, the weighting scheme promotes a more balanced learning objective. It encourages the model to maintain robust performance across the entire range of earthquake magnitudes, especially for the less frequent but more critical high-magnitude events.

3.5 Model Training and Validation

In machine learning, training is a fundamental process through which a model learns patterns and relationships from labeled data (e.g., Bishop 2006; Goodfellow et al. 2016). During this phase, the model iteratively adjusts its internal parameters to minimize a predefined loss function, thereby improving its ability to map input features to target outputs. This process is essential for enabling the model to capture the underlying structure of the data and develop predictive capability. However, optimizing the model solely on the training data may result in overfitting, where the model performs well on known data but fails to generalize to unseen examples. To mitigate this, a separate validation set is used during training to monitor the model's generalization performance. The validation process provides critical feedback, guiding model selection and hyperparameter tuning by indicating whether the learned representations extend beyond the training data.

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To enable rapid and accurate magnitude estimation from continuously updated real-time waveforms, we train a single unified model rather than multiple models. All waveform slices generated during the segmentation process are collectively incorporated into the same training dataset. Because each slice represents a waveform window of different duration, the model naturally learns a mapping that generalizes across variable input lengths, allowing it to operate effectively under the full range of real-time data availability encountered in EEWS operations. This design reflects the practical requirement that magnitude estimates must be continuously refined as additional waveform data become available. During inference, the model directly receives the waveform segment corresponding to the current EEWS report timing. Consequently, a single trained model is capable of producing reliable magnitude estimates for waveform inputs of varying durations, ensuring both computational efficiency and operational feasibility in real-time early warning scenarios.

In this study, the model is trained using the Adam optimizer (Kingma and Ba 2014), which combines the advantages of both AdaGrad and RMSProp by maintaining adaptive learning rates for each parameter and incorporating estimates of first- and second-order moments of the gradients. This enables efficient and stable convergence. The learning rate is set to 0.001, a value empirically determined to offer a balance between convergence speed and stability. The batch size is set to 512, which reflects a practical trade-off between memory efficiency and gradient estimation quality. Larger batches offer smoother gradient updates and benefit from GPU parallelism, while smaller batches may yield noisier updates and slower convergence. The chosen batch size was found to be effective for our dataset size and computational resources, ensuring efficient training without exceeding hardware limitations.

During each training epoch, batches of waveform and distance data are fed into the model, which computes predictions and evaluates the loss using the magnitude-weighted loss function. This loss is then backpropagated to compute gradients, which are used to update the model parameters (e.g., Goodfellow et al. 2016). One complete pass through the training dataset constitutes an epoch. After each epoch, model performance is evaluated on the validation dataset. If the validation loss is lower than all previous epochs, the current model weights are saved as the best-performing version. Otherwise, the weights remain unchanged. This training–validation cycle is repeated for a total of 200 epochs, allowing the model to progressively refine its ability to estimate earthquake magnitudes from waveform and distance inputs, although convergence is typically achieved within the first 100 epochs.

Model training and validation were conducted on a desktop workstation equipped with an AMD Ryzen 9 3950X processor and a single NVIDIA RTX 3090 GPU. The total training time was approximately 12 hours. Given the moderate computational requirements, the training and validation process remains feasible for most users equipped with

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commonly available hardware. This highlights the potential for researchers and institutions to build and fine-tune earthquake magnitude estimation models using their own datasets, without the need for high-performance computing infrastructure.

4. Results

To comprehensively evaluate the performance of the proposed model, we conducted a series of experiments across multiple aspects. First, we assessed the model using a dedicated test dataset that was not involved in the training or validation phases (e.g., Goodfellow et al. 2016). This allows us to objectively examine the model's generalization capability on previously unseen earthquake events. In addition, considering that the model is intended for real-time earthquake early warning applications, we evaluated its inference efficiency to ensure it can provide reliable magnitude estimates within the short timeframes. Finally, we applied the trained model to real-world earthquake early warning events recorded by CWA from January to June 2025, comparing the results with those obtained from the existing operational system. This comparison provides practical insights into the potential benefits and limitations of integrating machine learning-based magnitude estimation into future early warning frameworks.

4.1 Model Performance on the Test Dataset

To evaluate the generalization capability of the proposed model, we applied the trained network to a test dataset that was strictly held out during the training and validation processes. This allows us to assess the model's performance on previously unseen data, which is critical for validating its robustness and applicability to real-world seismic events. The test dataset includes earthquake records spanning a wide range of magnitudes, enabling us to further investigate the model's prediction performance across different magnitude intervals. To quantitatively assess model accuracy, we adopted three widely used statistical metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Standard Deviation (σ) of the prediction errors. These metrics were computed by comparing the predicted magnitudes with the catalog magnitudes for each test event. The MAE measures the average absolute difference between the predicted and true magnitudes, providing an intuitive indicator of overall prediction accuracy. In contrast, the RMSE gives greater weight to larger errors, making it more sensitive to outliers particularly useful for identifying occasional severe misestimations. The standard deviation (σ) of the prediction errors was also calculated as a supplementary metric to assess the stability and consistency of the model's predictions. A lower σ indicates that the model produces more uniform errors, reflecting a higher level of reliability across diverse events.

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Figure 4 illustrates the temporal evolution of model performance on the test dataset from 0.5 to 15 seconds after the fourth P-wave arrival. The three evaluation metrics (MAE, RMSE, and σ) of the prediction errors are plotted to assess how the model's estimation accuracy improves as more waveform data become available. All three metrics exhibit a clear downward trend over time, indicating that the model benefits from increasing information content in the waveform segments. Specifically, MAE decreases steadily from approximately 0.17 to below 0.11, demonstrating that the average prediction error becomes smaller as longer waveform windows are used. Similarly, RMSE, which is more sensitive to larger errors, also decreases and converges with the σ line in later stages, suggesting improved consistency and fewer large outliers in the predictions. The σ follows a trajectory similar to RMSE, further confirming that the model becomes more stable and less variable in its predictions as time progresses. These results collectively demonstrate that the model is capable of producing accurate and consistent magnitude estimates within a few seconds after the onset of P-wave arrivals, which is essential for timely earthquake early warning applications.

To further examine the temporal progression of the model's performance, we present a detailed comparison of predicted versus catalog magnitudes from 1 to 8 seconds after the fourth P-wave arrival (Figure 5). Each time step includes two subplots: a scatter plot comparing predicted and catalog magnitudes, and a corresponding histogram showing the distribution of prediction errors. In the scatter plots (Figure 5), most data points lie closely along the 1:1 reference line, indicating that the predicted magnitudes generally match the true magnitudes well. As the time window increases, the spread of data points around the diagonal line gradually narrows, showing improved prediction accuracy with longer waveform input.

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The histograms in Figure 5 further illustrate this trend. At $t = 1$ s, the error distribution is relatively wide, with an RMSE of 0.202 and an MAE of 0.153, indicating greater uncertainty in the earliest stage of prediction. From $t = 2$ s to $t = 8$ s, the error distribution becomes progressively more concentrated around zero, reflecting reduced bias and improved precision. By $t = 8$ s, the RMSE decreases to 0.145, the MAE to 0.111, and the standard deviation of errors shows a similar reduction. These results indicate that even within the first few seconds of waveform observation, the model is capable of producing reasonably accurate and stable magnitude estimates, with performance improving steadily as additional waveform data are incorporated.

It is noteworthy that the number of large-magnitude earthquakes in the dataset is relatively small; nevertheless, the model exhibits strong performance even for these less frequent events. This robustness is likely due to the adoption of a weighted loss function, which compensates for the imbalance in magnitude distribution, as well as the preservation of

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absolute waveform amplitudes by avoiding normalization (e.g., Mousavi and Beroza 2020; Hou et al. 2024). Retaining the original amplitude enables the model to more effectively capture the physical scaling of seismic signals, thereby enhancing its ability to accurately estimate the magnitudes of larger earthquakes.

4.2 Inference Speed and Computational Efficiency

To evaluate the real-time applicability of the proposed model, we measured its inference speed over 10,000 runs using an AMD Ryzen 9 3950X CPU on a desktop workstation. The resulting distribution of prediction times is shown in Figure 6. The average prediction time per waveform set was approximately 0.0230 seconds, with a standard deviation of only 0.0014 seconds, indicating a high level of computational stability and consistency. The performance results highlight one of the key strengths of the proposed model, its lightweight architecture. With a relatively small number of parameters and an efficient design, the model can generate predictions at near real-time speed, even without GPU acceleration. This efficiency is particularly advantageous for earthquake early warning (EEW) systems, where rapid decision-making is essential.

In an operational EEW scenario, once the seismic network detects a new event and an initial estimate of the epicenter and origin time is available, the proposed model can immediately process incoming waveform data and deliver a reliable magnitude estimate in just a few milliseconds. Such low-latency performance allows the model to serve as a critical component in the early phase of ground motion forecasting and hazard assessment, enhancing the system's ability to issue timely alerts. Overall, the results demonstrate that the proposed model achieves a favorable balance between accuracy and computational efficiency, making it a practical and scalable solution for real-world seismic monitoring applications.

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4.3 Application to Simulated Earthquake Early Warning Scenarios

To evaluate the effectiveness of the proposed model in a real-time operational setting, we applied it to a series of earthquake early warning (EEW) reports issued by the CWA between January and June 2025 (Figure 7). These reports were generated for earthquakes that triggered the EEWS during that period, offering a realistic testbed for assessing the model's practical performance. The CWA EEWS issues a sequence of updated reports over time as more waveform data become available. Among these, the third report typically contains the most reliable estimates of the source parameters of the event and is often used to support decision-making for issuing public alerts. However, even from the first report, the system already begins estimating the hypocenter and magnitude of the earthquake. Currently, the magnitude estimate is using the existing M_{pd} method, which is based on single-station predictions and statistical averaging (e.g., Chen et al. 2015, 2019).

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In our evaluation, we ensured a fair comparison by using the same waveform durations as those employed by the M_{pd} method at each report stage. For each EEW report, the proposed model used the corresponding waveform data and hypocenters to perform magnitude estimation. To further investigate the influence of source location accuracy, we conducted two sets of tests: The first used the hypocenters provided by the EEW report at each report stage. The second used more accurate hypocenters from the GDMS catalog, while keeping the waveform window consistent with that used in the M_{pd} evaluations. This dual testing approach allows us to assess the robustness of the model under typical operational uncertainties, while also demonstrating its potential under reliable hypocenter estimates. The results presented in Figure 8 clearly demonstrate the performance differences among the three tested approaches: the M_{pd} method currently used by the CWA, the proposed model using EEW-estimated hypocenters, and the proposed model using GDMS-based precise hypocenters. Across all three evaluation metrics, RMSE, MAE, and Standard Deviation (σ), the proposed model consistently outperforms the M_{pd} method. In all three panels (Figure 8a–c), the M_{pd} method exhibits the highest error values and the largest variability, particularly in the earlier reports. This is expected, as M_{pd} relies on per-station magnitude estimates and simple outlier rejection, which are prone to instability when data is sparse or when station coverage is uneven. Even as the EEW report sequence progresses from Report 1 to Report 5 and more waveform data becomes available, the M_{pd} performance, though improved, remains significantly inferior to the machine learning-based approaches.

The proposed model, when applied with EEW-derived hypocenters, achieves substantially improved accuracy and stability compared to M_{Pd}, starting from the first report and persisting consistently through the fifth (Figure 8). This result highlights the robustness of the model, even when the input hypocenter estimates include the uncertainties that are typical of early-stage EEW. Moreover, when the model is supplied with more accurate hypocentral information from the GDMS catalog (Figure 8), its performance improves even further. The error metrics across all report stages are the lowest in this setting, with RMSE consistently below 0.15 and standard deviation values near 0.12. This result confirms that, in addition to waveform content, the quality of hypocenter inputs is a critical factor influencing magnitude estimation performance.

Overall, these findings demonstrate that the proposed model provides not only more accurate but also more consistent and reliable magnitude estimates compared to the existing M_{Pd} method (Figure 8). Its ability to perform effectively even with early and uncertain inputs highlights the model's strong potential for operational deployment in real-time EEWS.

5.Discussion

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5.1 Effect of Magnitude-Weighted Loss on Early Magnitude Estimation

To evaluate the role of loss-function design in earthquake magnitude estimation, we compared model performance using weighted and unweighted loss formulations. The two models were trained using identical datasets, network architectures, and training procedures, ensuring that any observed differences can be primarily attributed to the choice of loss function.

We first evaluated model performance for earthquakes with magnitudes exceeding 5.0, which correspond to the operational warning threshold of the Central Weather Administration (CWA). Overall, the weighted-loss model demonstrates consistently lower error metrics and reduced prediction variability across most time windows (Figure 9a–c). Although the observed improvements are modest and do not always reach statistical significance, the results suggest that the weighted formulation contributes to more stable and reliable magnitude estimates for high-impact events during the early stages of waveform accumulation. Such incremental improvements are operationally meaningful in time-critical earthquake early warning scenarios. When evaluated across the full test dataset, the performance of the weighted and unweighted models remains largely comparable, indicating that the introduction of magnitude weighting does not compromise estimation accuracy for the more abundant small-to-moderate earthquakes. Notably, the weighted-loss model exhibits slightly improved performance during intermediate time windows (approximately 3–9 s after P-wave arrival; Figure 9d–f).

It is important to note that the number of large-magnitude events available for evaluation remains limited, accounting for approximately 6.1% of the test dataset. This data limitation likely constrains the statistical significance of the observed differences between the weighted and unweighted formulations. As additional large-magnitude earthquakes are recorded in future operations, the effectiveness of magnitude-weighted training can be assessed more rigorously. With an expanded dataset that includes a greater number of large-magnitude events, the benefits of the weighted loss function may become more pronounced, further improving the robustness and reliability of early magnitude estimates in operational earthquake early warning systems.

5.2 Comparisons of FNO, CNN, and FC Architectures

To evaluate the effectiveness of the Fourier Neural Operator (FNO) in extracting seismic waveform features for earthquake magnitude estimation, we conducted a systematic comparison with two widely used baseline architectures: Convolutional Neural Networks (CNN) and Fully Connected (FC) layers (e.g., Bishop 2006; Goodfellow et al. 2016). In this experiment, the FNO–based feature-extraction backbone was systematically replaced with either a CNN or a FC architecture (Table 1),

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721 while all other model components and training configurations were kept
722 identical. Specifically, all three model variants employ the same input
723 representation, LSTM–based temporal modeling block, and fully
724 connected output projection layer. It is noted that the three architectures
725 differ substantially in model complexity: the FC-based model contains on
726 the order of 10^6 trainable parameters, whereas the CNN- and FNO-based
727 models involve on the order of 10^5 parameters. Consequently, the
728 architectural comparison spans more than one order of magnitude in
729 parameter count, providing a basis for assessing the efficiency of
730 different feature-extraction strategies under a controlled experimental
731 setting. To ensure a fair and controlled comparison, the CNN and FC
732 models were trained using the same training, validation, and test
733 datasets as the FNO model. The training and validation parameters were
734 also identical across all models, consistent with the setup described in
735 Section 3.5. This design minimizes confounding factors, ensuring that
736 performance differences reflect the intrinsic capabilities of the model
737 architectures rather than variations in data or training procedures.

738 The comparison results are presented in terms of RMSE, MAE, and the
739 standard deviation (σ) of prediction errors (Figure 10), evaluated as a
740 function of time after the fourth triggered station. Across all three
741 metrics, the FNO-based model consistently outperforms the CNN and FC
742 models, particularly during the early time window (1–6 seconds), which is
743 critical for real-time earthquake early warning applications. The FNO
744 model not only achieves lower prediction errors but also exhibits reduced
745 uncertainty, as evidenced by the consistently smaller standard deviations
746 of its outputs. This superior performance can be attributed to the model's
747 ability to learn waveform features in the frequency domain more
748 effectively than local (CNN) or dense (FC) architectures. These findings
749 support the adoption of FNO-based architectures as a reliable approach
750 for rapid and reliable earthquake magnitude estimation, especially in
751 scenarios where only short time windows of waveform data are available.

752

753 **5.3 Improving Magnitude Estimation Through Enhanced**
754 **Hypocenter Accuracy**

755 Accurate hypocenter estimation is a critical factor influencing the
756 reliability of earthquake magnitude determination. In our evaluation of
757 magnitudes derived from CWA EEWS reports, we observed that
758 improved hypocenter accuracy substantially reduces uncertainty and
759 enhances estimation precision. As shown in Figure 6, magnitude
760 estimates become more reliable when refined hypocenters from the
761 GDMS catalog are used. At present, the CWA EEWS system employs a
762 simplified and computationally efficient 1D velocity model for rapid
763 hypocenter estimation. Although this approach enables real-time
764 operation, it can introduce non-negligible location errors, particularly in
765 geologically complex regions such as Taiwan.

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To further quantify the sensitivity of the proposed magnitude-estimation framework to hypocentral location uncertainty, we conducted a systematic evaluation by introducing synthetic location errors of 5 km, 10 km, and 20 km. Figure 11 illustrates the resulting magnitude estimation performance as a function of available waveform duration. The results reveal a clear dependence of magnitude accuracy on hypocentral error: increasing location errors lead to systematically larger RMSE and MAE values, as well as increased prediction variability (σ), across all time windows. Notably, the degradation in performance is most pronounced during the early stages of waveform accumulation, when limited signal information is available and the estimation is more sensitive to geometric factors such as epicentral distance. As the waveform duration increases, the impact of hypocentral error is partially mitigated, but even at later times, larger location errors continue to impose a measurable penalty on magnitude accuracy. Overall, this analysis highlights the importance of improving hypocenter accuracy in operational earthquake early warning systems. While the proposed magnitude-estimation model demonstrates robustness to moderate location errors, enhanced hypocentral constraints can improve the reliability and stability of early magnitude estimates.

Recent developments, such as the study by Lian et al. (2024), have demonstrated that integrating machine learning with a 3D velocity model–based equal differential time (EDT) grid search method can significantly improve the accuracy of near real-time hypocenter estimates. This method employs a grid-based search using a precomputed travel-time table derived from a 3D velocity model, enabling both high accuracy and computational efficiency in hypocenter determination. Furthermore, it can be combined with real-time P-wave picking algorithms, such as RED-PAN, to ensure that reliable seismic arrivals are incorporated into the location and magnitude estimation processes. Improved hypocenter accuracy contributes to more precise magnitude estimation, which is especially critical in the early stages of seismic events to ensure the effective operation of EEWS. When combined with robust magnitude estimation models, such as those developed in this study, the overall reliability of EEWS can be further strengthened. Importantly, achieving relatively reliable magnitude estimates as early as the first report could reduce the reliance on later updates (e.g., the third report) for confident decision-making in Taiwan's EEWS (Figure 8). This advancement has the potential to shorten the decision-making timeline, reduce the lead time for issuing alerts, and ultimately increase public trust in the system while enabling more timely protective actions during seismic emergencies.

6. Conclusion

This study presents a frequency-domain multi-station deep learning framework that integrates FNO layers with LSTM networks to enable rapid and robust earthquake magnitude estimation for Taiwan's EEWS.

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By incorporating multi-component waveform data from multiple stations alongside hypocentral distance corrections, the model effectively captures both spectral and temporal features that are critical for accurate magnitude estimation. Comprehensive evaluations demonstrate that the proposed method consistently outperforms the existing Pd-based single-station approach currently employed by the CWA (Chen et al. 2015). The performance gains are particularly notable during the earliest stages of EEWS reporting, providing more accurate magnitude assessments within just a few seconds after the fourth P-wave arrival. Even when operating under the inherent location uncertainties of early EEWS stages, the model produces stable, low-variance predictions (Figure 8). When supplemented with precise hypocenter information, such as that from the GDMS catalog, the accuracy improves further. Moreover, integrating 3-D velocity models with machine learning–based seismic phase picking enables rapid and accurate hypocenter refinement (Lian et al. 2024), which in turn reduces source location uncertainties, enhances the reliability of initial magnitude estimates, and improves the overall effectiveness of EEWS.

In addition to its accuracy benefits, the proposed model is computationally efficient, capable of producing predictions within milliseconds on standard CPUs. This efficiency makes it well-suited for real-time operational deployment without imposing substantial hardware requirements. The capability to seamlessly integrate into existing EEWS pipelines offers the potential to shorten decision-making timelines, increase warning lead times, and ultimately strengthen public safety and disaster mitigation efforts in Taiwan. By combining advanced machine learning techniques with dense seismic observations, this research offers a pathway toward faster, more accurate, and more reliable earthquake early warning systems, providing crucial seconds that can save lives and reduce economic losses in earthquake-prone regions such as Taiwan.

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References

Allen RM, Kanamori H (2003) The Potential for Earthquake Early Warning in Southern California. Science 300:786–789. https://doi.org/10.1126/science.1080912

ARTICLE IN PRESS

856 Allen RM, Melgar D (2019) Earthquake Early Warning: Advances,
857 Scientific Challenges, and Societal Needs. Annu Rev Earth Planet
858 Sci 47:361–388. https://doi.org/10.1146/annurev-earth-053018-
859 060457

860 Bishop CM (2006) Pattern Recognition and Machine Learning. New York

861 Byrne T, Chan Y-C, Rau R-J, et al (2011) The Arc–Continent Collision in
862 Taiwan. In: Arc-Continent Collision. Springer Berlin Heidelberg,
863 Berlin, Heidelberg, pp 213–245

864 Chen D, Hsiao N, Wu Y (2015) The Earthworm Based Earthquake Alarm
865 Reporting System in Taiwan. Bulletin of the Seismological Society
866 of America 105:568–579. https://doi.org/10.1785/0120140147

867 Chen D-Y, Lin T-L, Hsu H-C, et al (2019) An approach to improve the
868 performance of the earthquake early warning system for the 2018
869 Hualien earthquake in Taiwan. Terr Atmos Ocean Sci 30:423–433.
870 https://doi.org/10.3319/TAO.2018.12.23.02

871 Chen D-Y, Wu Y-M, Chin T-L (2017) An empirical evolutionary magnitude
872 estimation for early warning of earthquakes. Journal of Asian Earth
873 Sciences 135:190–197. https://doi.org/10.1016/j.jseaes.2016.12.028

874 Elkan C (2001) The foundations of cost-sensitive learning. In:
875 Proceedings of the 17th international joint conference on Artificial
876 intelligence - Volume 2. Morgan Kaufmann Publishers Inc., San
877 Francisco, CA, USA, pp 973–978

878 Gasparini P, Manfredi G, Zschau J (eds) (2007) Earthquake Early
879 Warning Systems. Springer, Berlin, Heidelberg

880 Gers FA, Schmidhuber J, Cummins F (2000) Learning to Forget:
881 Continual Prediction with LSTM. Neural Computation 12:2451–
882 2471. https://doi.org/10.1162/089976600300015015

883 Goodfellow I, Bengio Y, Courville A, Bengio Y (2016) Deep learning. MIT
884 press Cambridge

885 He H, Garcia EA (2009) Learning from Imbalanced Data. IEEE
886 Transactions on Knowledge and Data Engineering 21:1263–1284.
887 https://doi.org/10.1109/TKDE.2008.239

888 Ho CS (1986) A synthesis of the geologic evolution of Taiwan.
889 Tectonophysics 125:1–16. https://doi.org/10.1016/0040-
890 1951(86)90004-1

891 Hochreiter S, Schmidhuber J (1997) Long Short-Term Memory. Neural
892 Computation 9:1735–1780.
893 https://doi.org/10.1162/neco.1997.9.8.1735

ARTICLE IN PRESS

894 Hou B, Zhou Y, Li S, et al (2024) Real-time earthquake magnitude
895 estimation via a deep learning network based on waveform and text
896 mixed modal. Earth Planets Space 76:58.
897 https://doi.org/10.1186/s40623-024-02005-8

898 Huang C-Y, Yuan PB, Lin C-W, et al (2000) Geodynamic processes of
899 Taiwan arc–continent collision and comparison with analogs in
900 Timor, Papua New Guinea, Urals and Corsica. Tectonophysics
901 325:1–21

902 Jenkins WF, Gerstoft P, Bianco MJ, Bromirski PD (2021) Unsupervised
903 Deep Clustering of Seismic Data: Monitoring the Ross Ice Shelf,
904 Antarctica. J Geophys Res Solid Earth 126:.
905 https://doi.org/10.1029/2021JB021716

906 Joshi A, Vishnu C, Mohan CK, Raman B (2023) Application of XGBoost
907 model for early prediction of earthquake magnitude from waveform
908 data. J Earth Syst Sci 133:5. https://doi.org/10.1007/s12040-023-
909 02210-1

910 Jozefowicz R, Zaremba W, Sutskever I (2015) An empirical exploration of
911 recurrent network architectures. In: Proceedings of the 32nd
912 International Conference on International Conference on Machine
913 Learning - Volume 37. JMLR.org, Lille, France, pp 2342–2350

914 Kingma DP, Ba J (2014) Adam: A method for stochastic optimization.
915 arXiv preprint arXiv:14126980

916 Krawczyk B (2016) Learning from imbalanced data: open challenges and
917 future directions. Prog Artif Intell 5:221–232.
918 https://doi.org/10.1007/s13748-016-0094-0

919 Lara P, Bletery Q, Ampuero J-P, et al (2023) Earthquake Early Warning
920 Starting From 3 s of Records on a Single Station With Machine
921 Learning. Journal of Geophysical Research: Solid Earth
922 128:e2023JB026575. https://doi.org/10.1029/2023JB026575

923 Li Z, Kovachki N, Azizzadenesheli K, et al (2021) Fourier Neural
924 Operator for Parametric Partial Differential Equations

925 Lian J-X, Liao W-Y, Lee E-J, et al (2024) Integration of Machine learning
926 and equal differential time method for enhanced hypocenter
927 localization in earthquake early warning systems: application to
928 dense seismic arrays in Taiwan. Earth, Planets and Space 76:94.
929 https://doi.org/10.1186/s40623-024-02037-0

930 Liao W-Y, Lee E-J, Chen D-Y, et al (2022) RED-PAN: Real-Time
931 Earthquake Detection and Phase-Picking With Multitask Attention
932 Network. IEEE Transactions on Geoscience and Remote Sensing
933 60:1–11. https://doi.org/10.1109/TGRS.2022.3205558

ARTICLE IN PRESS

934 Liao W-Y, Lee E-J, Mu D, et al (2021) ARRU Phase Picker: Attention
935 Recurrent-Residual U-Net for Picking Seismic P- and S-Phase
936 Arrivals. Seismological Research Letters 92:2410–2428.
937 https://doi.org/10.1785/0220200382

938 Liu K-S, Shin T-C, Tsai Y-B (1999) A free-field strong motion network in
939 Taiwan: TSMIP. Terrestrial, Atmospheric and Oceanic Sciences
940 10:377–396

941 Mousavi SM, Beroza GC (2020) A Machine-Learning Approach for
942 Earthquake Magnitude Estimation. Geophysical Research Letters
943 47:. https://doi.org/10.1029/2019GL085976

944 Mousavi SM, Ellsworth WL, Zhu W, et al (2020) Earthquake
945 transformer—an attentive deep-learning model for simultaneous
946 earthquake detection and phase picking. Nature Communications
947 11:. https://doi.org/10.1038/s41467-020-17591-w

948 Mousavi SM, Sheng Y, Zhu W, Beroza GC (2019) STanford EArthquake
949 Dataset (STEAD): A Global Data Set of Seismic Signals for AI. IEEE
950 Access 7:179464–179476.
951 https://doi.org/10.1109/ACCESS.2019.2947848

952 Münchmeyer J, Bindi D, Leser U, Tilmann F (2021) Earthquake
953 magnitude and location estimation from real time seismic
954 waveforms with a transformer network. Geophysical Journal
955 International 226:1086–1104. https://doi.org/10.1093/gji/ggab139

956 Navarro-Rodríguez A, Castro-Artola OA, García-Guerrero EE, et al (2025)
957 Recent Advances in Early Earthquake Magnitude Estimation by
958 Using Machine Learning Algorithms: A Systematic Review. Applied
959 Sciences 15:3492. https://doi.org/10.3390/app15073492

960 Saad OM, Chen Y, Savvaidis A, et al (2022) Real-Time Earthquake
961 Detection and Magnitude Estimation Using Vision Transformer.
962 JGR Solid Earth 127:. https://doi.org/10.1029/2021JB023657

963 Seydoux L, Balestriero R, Poli P, et al (2020) Clustering earthquake
964 signals and background noises in continuous seismic data with
965 unsupervised deep learning. Nat Commun 11:3972.
966 https://doi.org/10.1038/s41467-020-17841-x

967 Song J, Zhu J, Li S (2022) MEANet: Magnitude estimation via physics-
968 based features time series, an attention mechanism, and neural
969 networks. Geophysics 88:V33–V43.
970 https://doi.org/10.1190/geo2022-0196.1

971 Sun H, Ross ZE, Zhu W, Azizzadenesheli K (2023) Phase Neural Operator
972 for Multi-Station Picking of Seismic Arrivals. Geophysical Research
973 Letters 50:e2023GL106434. https://doi.org/10.1029/2023GL106434

ARTICLE IN PRESS

ARTICLE IN PRESS

Teng LS (1990) Geotectonic evolution of late Cenozoic arc-continent collision in Taiwan. Tectonophysics 183:57–76

van den Ende MPA, Ampuero J-P (2020) Automated Seismic Source Characterization Using Deep Graph Neural Networks. Geophysical Research Letters 47:e2020GL088690. https://doi.org/10.1029/2020GL088690

Wang C, Lee E, Liao W, et al (2023) Cluster Analysis of Slope Hazard Seismic Recordings Based Upon Unsupervised Deep Embedded Clustering. Seismological Research Letters. https://doi.org/10.1785/0220230011

Wu Y-M, Hsiao N-C, Lee WHK, et al (2007a) State of the Art and Progress in the Earthquake Early Warning System in Taiwan. In: Gasparini P, Manfredi G, Zschau J (eds) Earthquake Early Warning Systems. Springer, Berlin, Heidelberg, pp 283–306

Wu Y-M, Kanamori H (2008) Development of an Earthquake Early Warning System Using Real-Time Strong Motion Signals. Sensors 8:1–9. https://doi.org/10.3390/s8010001

Wu Y-M, Kanamori H, Allen RM, Hauksson E (2007b) Determination of earthquake early warning parameters, τ_c and P_d, for southern California. Geophysical Journal International 170:711–717. https://doi.org/10.1111/j.1365-246X.2007.03430.x

Wu Y-M, Zhao L (2006) Magnitude estimation using the first three seconds P-wave amplitude in earthquake early warning. Geophysical Research Letters 33:. https://doi.org/10.1029/2006GL026871

Zhang X, Zhang M, Tian X (2021) Real-Time Earthquake Early Warning With Deep Learning: Application to the 2016 M 6.0 Central Apennines, Italy Earthquake. Geophysical Research Letters 48:2020GL089394. https://doi.org/10.1029/2020GL089394

Zhu J, Li S, Song J (2021) Magnitude Estimation for Earthquake Early Warning with Multiple Parameter Inputs and a Support Vector Machine. Seismological Research Letters 93:126–136. https://doi.org/10.1785/0220210144

Zhu W, Beroza GC (2018) PhaseNet: A Deep-Neural-Network-Based Seismic Arrival Time Picking Method. Geophysical Journal International. https://doi.org/10.1093/gji/ggy423

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Table

Table 1. Architectural comparison of the three model variants evaluated in this study, including the fully connected (FC), convolutional neural network (CNN), and Fourier Neural Operator (FNO)–based models.

Component	FC	CNN	FNO
Number of parameters	1,894,557	207,540	259,383
Input	3 × 10 × 2000 (channels × stations × samples)		
Feature extraction	3 fully connected layers	4 convolutional blocks (convolution + max pooling)	3 Fourier Neural Operator (FNO) layers
Temporal modeling	Long Short-Term Memory (LSTM) block		
Output projection	Fully connected output layer		
Output	1 (magnitude)		

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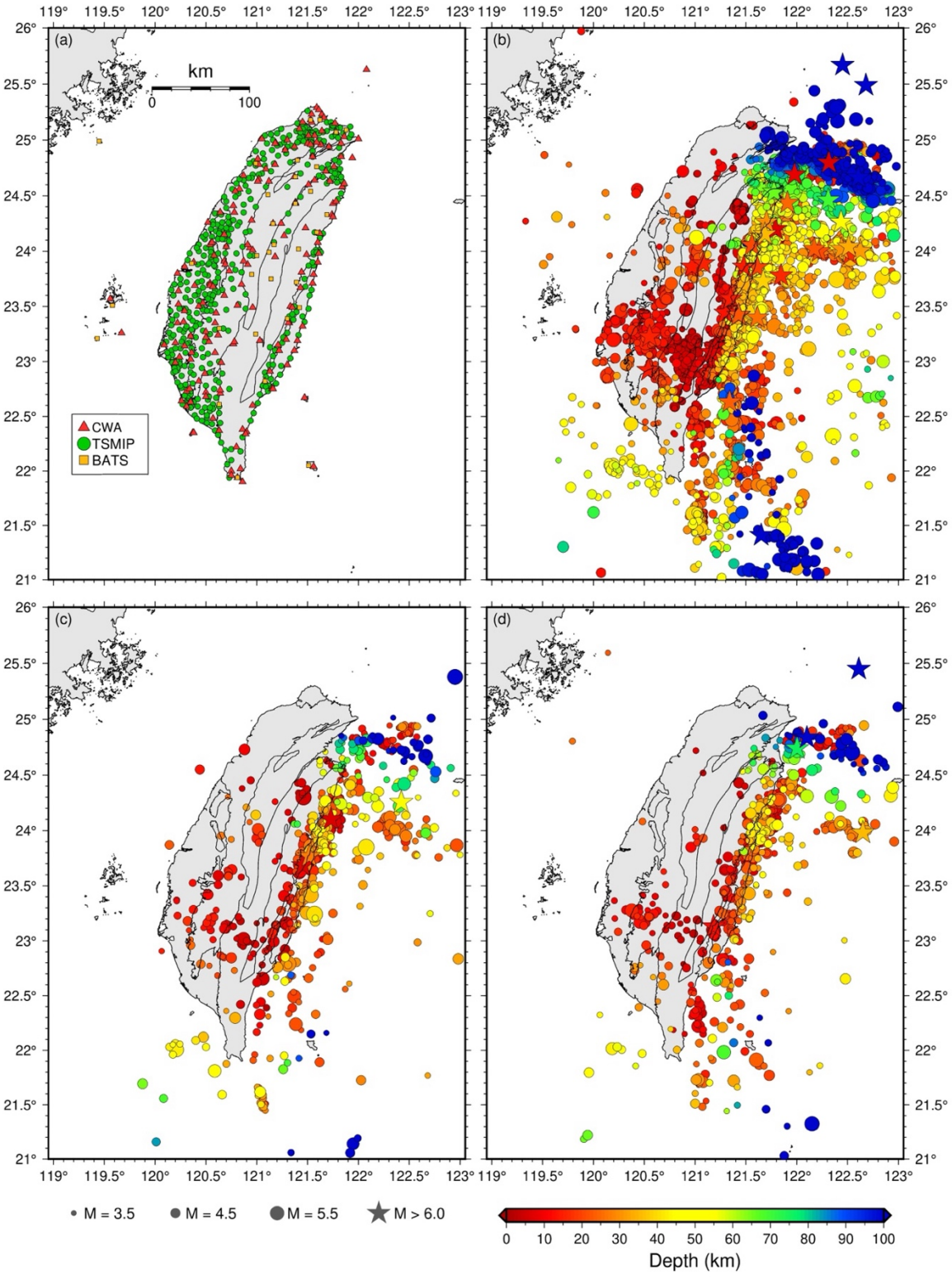


Figure 1. Distribution of seismic stations and earthquake events used in this study. (a) Seismic station networks, including the Central Weather Administration Seismic Network (CWASN, red triangles), the Taiwan Strong Motion Instrumentation Program (TSMIP, green circles), and the Broadband Array in Taiwan for Seismology (BATS, yellow squares). (b) Spatial distribution of earthquake events used for training, (c) validation,

and (d) testing of the proposed model. Event colors indicate focal depths, while symbol sizes scale with earthquake magnitude.

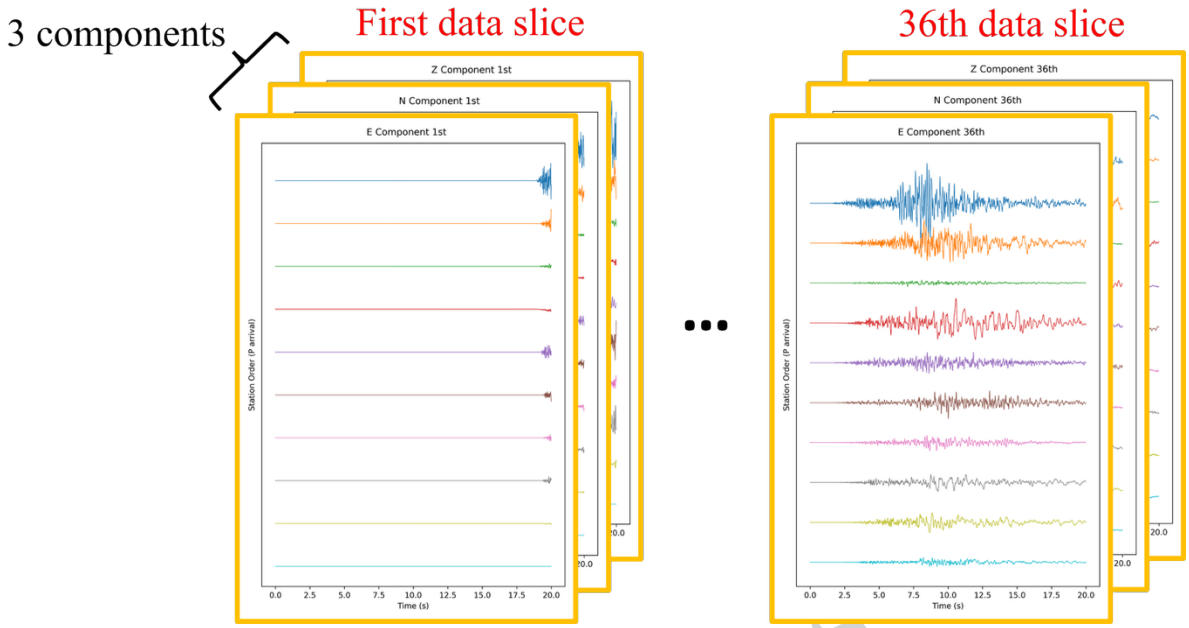


Figure 2. Schematic illustration of the input waveform segmentation. For each earthquake event, three-component recordings (E, N, and Z) from the ten nearest stations are selected, with each trace retaining a length of 20 seconds. To simulate real-time conditions, the waveforms are shifted in time and divided into sequential slices. The figure shows examples of the first and the 36th data slices.

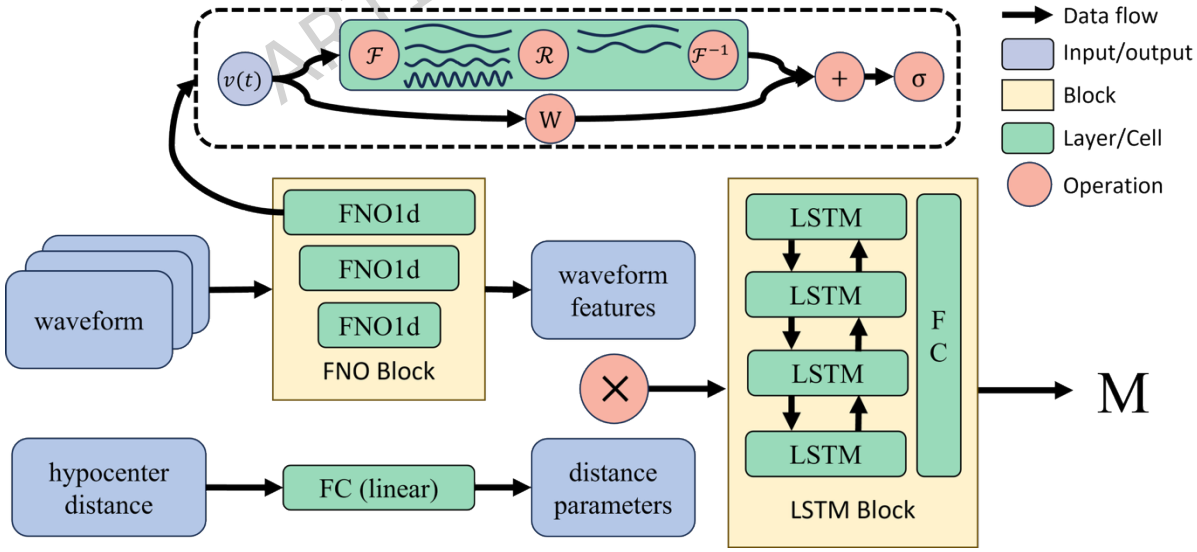


Figure 3. Schematic architecture of the proposed model for earthquake magnitude estimation. The model takes three-component waveform data from multiple stations as input and processes them through a Fourier Neural Operator (FNO) block to extract spectral-temporal features. In parallel, hypocentral distance information is transformed through a fully

connected (FC) layer to generate distance-related parameters. These two feature sets are combined and subsequently passed into a Long Short-Term Memory (LSTM) block to capture temporal dependencies in the waveform features. The final output is the predicted earthquake magnitude (M). The inset illustrates the internal structure of the FNO block, which includes Fourier transform (F), inverse Fourier transform (F^{-1}), residual connection (R), and weight operator (W).

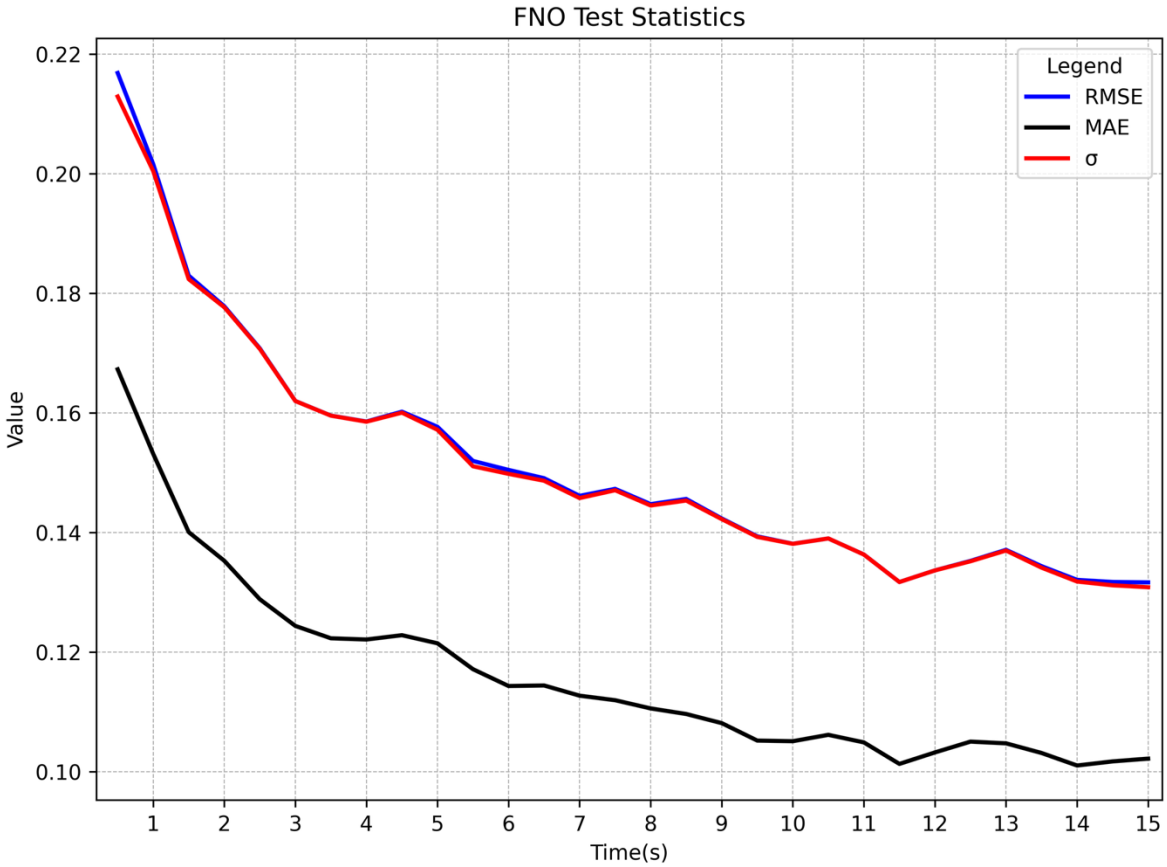


Figure 4. Temporal performance statistics of the proposed model on the test dataset. The figure shows the Mean Absolute Error (MAE, blue), Root Mean Squared Error (RMSE, blue), and Standard Deviation (σ, red) of the magnitude prediction errors from 0.5 to 15 seconds after the fourth P-wave arrival. All metrics improve (i.e., decrease) as time increases, indicating enhanced accuracy and stability with longer waveform input durations.

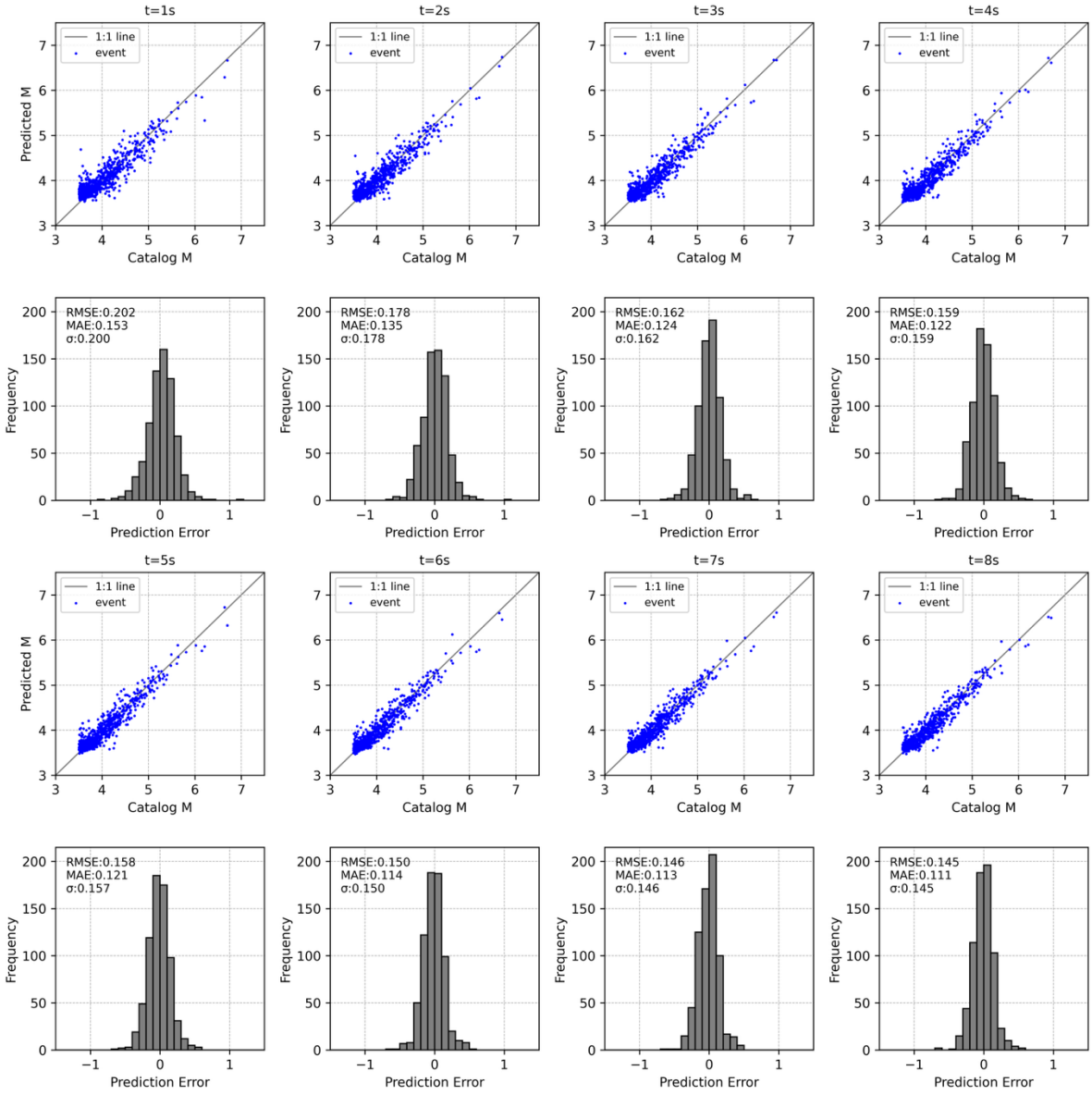


Figure 5. Scatter plots and error histograms of the proposed model evaluated on the test dataset at different time steps (1–8 seconds) after the fourth P-wave arrival. The first and third rows present predicted versus catalog magnitudes, while the second and fourth rows show the corresponding distributions of prediction errors. Each histogram reports root mean square error (RMSE), mean absolute error (MAE), and the standard deviation (σ) of prediction errors. The results indicate that the model achieves progressively higher accuracy and reduced variance as more waveform data become available over time.

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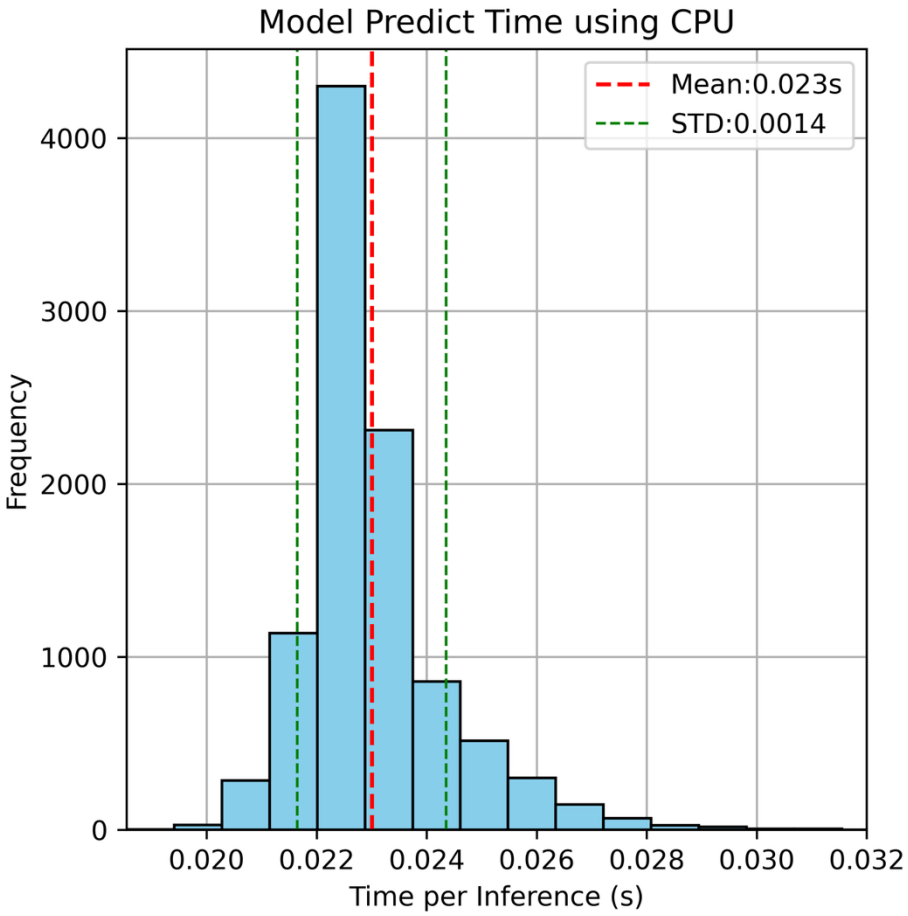


Figure 6. Distribution of model prediction times measured over 10,000 runs on an AMD Ryzen 9 3950X CPU. The average inference time is approximately 0.023 seconds, with a standard deviation of 0.0014 seconds.

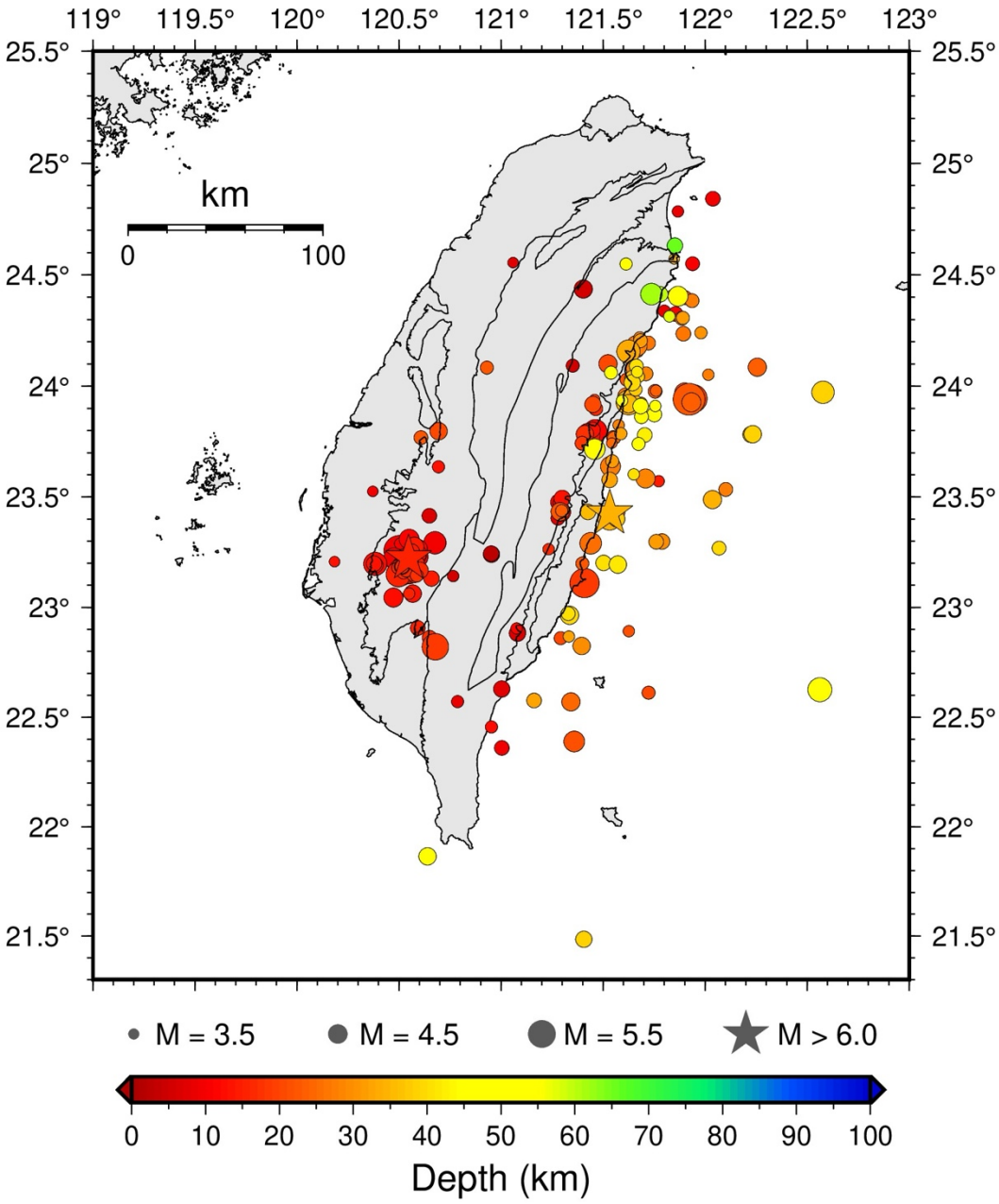


Figure 7. Spatial distribution of test earthquake events issued by the Central Weather Administration (CWA) Earthquake Early Warning System (EEWS) between January and June 2025. Symbol sizes scale with earthquake magnitude. Event colors represent focal depths.

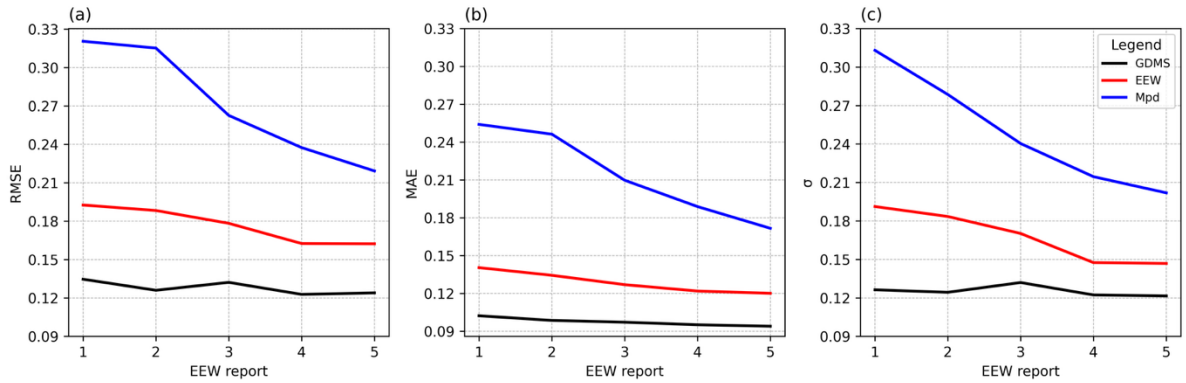


Figure 8. Comparison of earthquake magnitude estimation performance across different CWA EEWS reports (from the first to the fifth) using three approaches: the proposed model with EEWS-derived hypocenters (red), the proposed model with GDMS catalog hypocenters (black), and the M_{Pd}-based estimates from the CWA EEWS reports (blue). Panels show (a) root mean square error (RMSE), (b) mean absolute error (MAE), and (c) standard deviation (σ) of prediction errors. The results indicate that the proposed model consistently outperforms the M_{Pd}-based estimates, with performance further enhanced when accurate hypocenters from the GDMS catalog are employed.

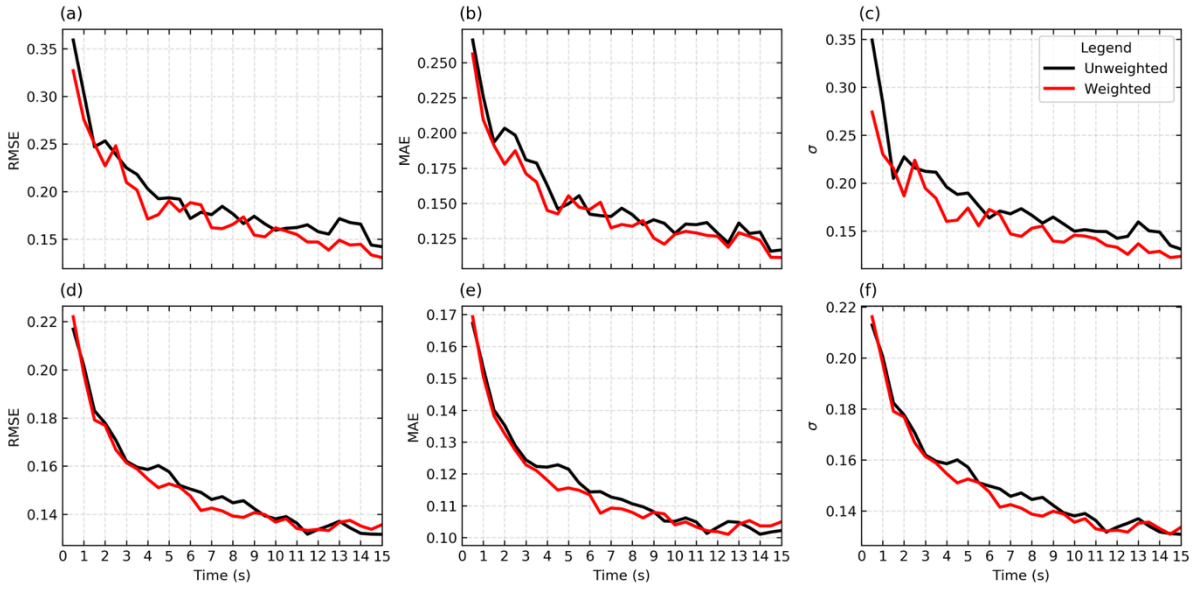


Figure 9. Comparison of magnitude estimation performance using weighted and unweighted loss functions. Panels (a)–(c) show the root mean square error (RMSE), mean absolute error (MAE), and standard deviation (σ) for earthquakes with magnitudes M_L > 5.0. Panels (d)–(f) present the same metrics for all events in the test dataset. Black and red curves denote results obtained using unweighted and magnitude-weighted loss functions, respectively.

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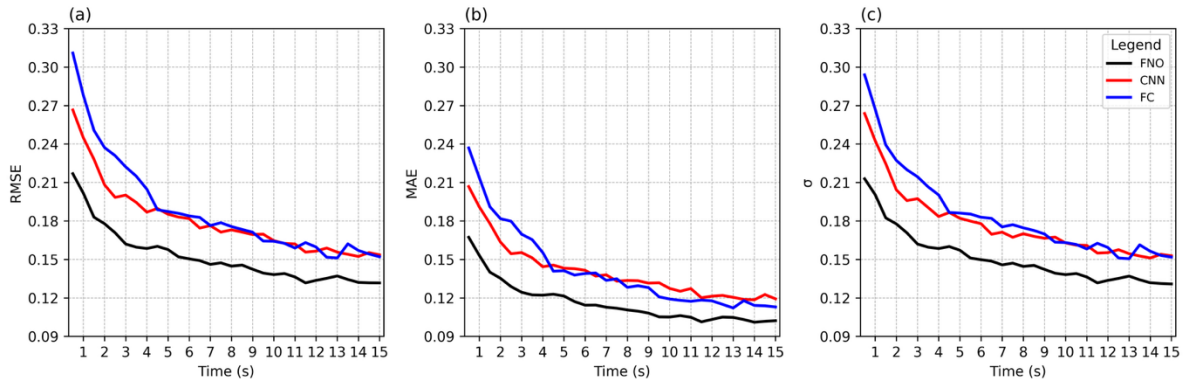


Figure 10. Performance comparison of three models, Fourier Neural Operator (FNO, black), Convolutional Neural Network (CNN, red), and Fully Connected network (FC, blue), evaluated on the test dataset. Panels show the evolution of (a) root mean square error (RMSE), (b) mean absolute error (MAE), and (c) standard deviation (σ) of prediction errors as a function of time since the fourth P-wave arrival. Results demonstrate that the FNO-based model consistently outperforms the CNN and FC models across all metrics, particularly during the earliest stages of prediction.

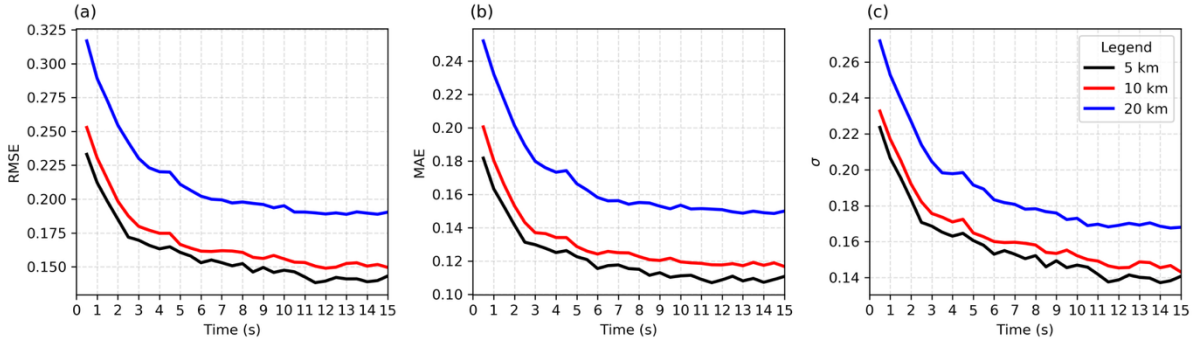


Figure 11. Impact of hypocentral location errors on magnitude estimation performance. Panels (a)–(c) show the root mean square error (RMSE), mean absolute error (MAE), and standard deviation (σ), respectively, evaluated under assumed hypocentral location errors of 5 km, 10 km, and 20 km.